MoleQuery

AI-Assisted Compound Research & Knowledge Management Platform

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A centralized web platform that unifies compound data, research documents, and AI-powered semantic search — helping researchers find and understand scientific information faster, using Retrieval-Augmented Generation (RAG).

The Problem & Why It Matters

Current Challenges

- Compound data is scattered across papers, PDFs, and spreadsheets
- Manual document search wastes researcher time
- Compound–target relationships are hard to track
- Keyword search misses semantically relevant results

Business Impact

- Longer research cycles and repeated document reading
- Missed or duplicated research effort
- Poorly organized institutional knowledge
- Documents stay "dead weight" instead of usable insight

Core Capabilities

🧪 Compound Knowledge

- Structured compound records: names, synonyms, formulas, descriptions
- Link compounds ↔ biological targets ↔ categories
- Search, filter & compare compounds

🗂 Document Management

- Upload PDF, DOCX & TXT research documents
- Automatic text extraction and chunking
- Centralized, searchable knowledge base

🤖 AI Research Assistant

- Converts natural-language questions into embeddings
- Retrieves the most relevant research context (RAG)
- Generates source-grounded answers + query history

How the AI (RAG) Engine Works

1. **Upload Document** →
2. **Extract & Chunk Text** →
3. **Generate Embeddings** →
4. **Ask a Question** →
5. **Retrieve Relevant Chunks** →
6. **AI Generates Answer**

Technology Stack

Frontend: React.js + Vite · Material UI · Axios

Backend: ASP.NET Core Web API · Entity Framework Core · JWT Auth

AI / Data: Embeddings · Vector Search · RAG

Outcomes & What's Next

Delivered Today

- Centralized, structured research knowledge base
- Faster, semantic information retrieval
- AI-assisted Q&A with source-aware responses
- Role-based access control

Future Roadmap

- Automated literature ingestion & larger knowledge bases
- Advanced molecular comparison & evidence ranking
- Research-topic recommendations & exportable summaries

Bottom line: MoleQuery turns scattered compound data and research papers into one searchable, AI-queryable knowledge system — cutting research time and improving evidence discovery.